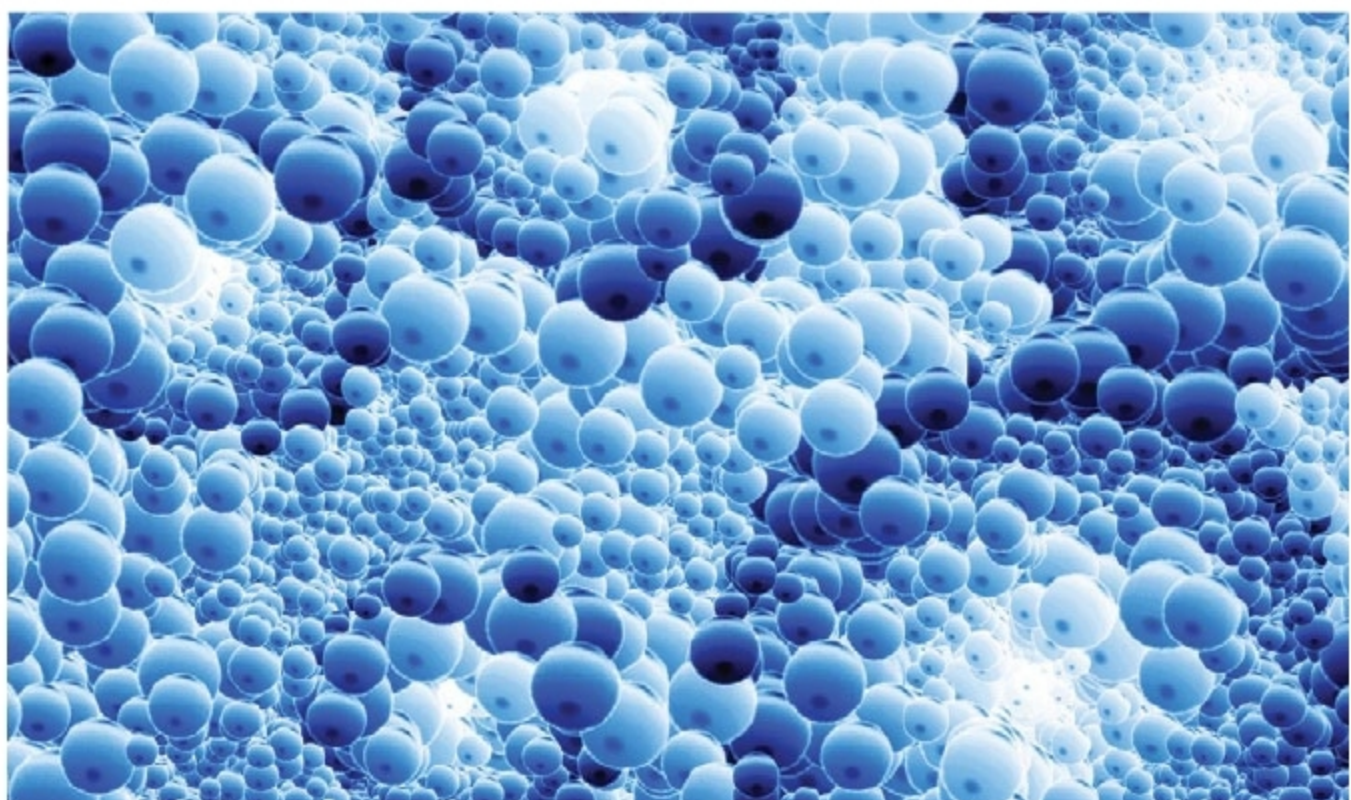


Exploring The Applications Of NANOTECHNOLOGY And NANOMATERIALS

In simple terms, nanotechnology is the part of science that deals with the control of matter with dimensions smaller than 100 nanometres, and can go down to atomic and molecular scales

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The study and manipulation of matter, particles and structures on the nanometer scale is referred to as nanoscience. Nanotechnology is the application of nanoscience that deals with the use of nanomaterials and nano-sized components in various products. It provides the ability to design custom-made materials and products with new and enhanced properties. It has a wide range of applications in nanoelectronic components, medicines and sensors, and interfaces between electronics and biological systems.

Scientists and engineers are deliberately making nanomaterials to take advantage of their enhanced properties, such as lighter weight, higher strength, increased control of light spectrum and greater chemical reactivity. Majority of biological processes occur at the nanoscale, including essential molecular reactions within the human body and components of foods. Dealing at the nanoscale enables scientists to utilise the unique physical, mechanical, chemical and optical properties of materials.

Need for development of nanomaterials

Nanomaterials have a high surface-to-volume ratio than large-scale materials of similar masses. As surface area increases, a greater amount of the material can come into contact, thus affecting reactivity. Greater surface area and improved reactivity help create better catalysts. Catalysis by engineered nanostructured materials has impacted the

catalyst market globally. For example, a catalytic converter in a car reduces the toxicity of the engine's fumes.

Nanoengineered batteries, fuel cells and catalysts produce cleaner and safer modes of storing energy. A large surface area makes nanomaterials suitable for water treatment and desalination. It supports applications ranging from drug delivery to clothing insulation. Using nanotechnology, materials can effectively be made stronger, lighter, more durable, more reactive or better electrical conductors, among many other traits.

Poonam Sharma, research fellow at The Energy and Resources Institute (TERI), comments, "The need to develop nanoparticles is different based on their applications, as we are utilising nano in various sectors. An example is micro-electro-mechanical systems (MEMS) wherein we shifted to nanoelectromechanical systems (NEMS), which helps in enhancements in properties and performance. Plenty of instruments are utilised to study nanoparticles, and it all depends on the kind of property you are looking for and the kind of nanoparticle it is."

Nanomaterials of different dimensions

Nanomaterials can be broadly classified as one-dimensional (1D), two-dimensional (2D) and three-dimensional (3D). Nanowire nanomaterials are examples of 1D nanomaterials, which improve both energy capacity and charge rate in rechargeable batteries. Activated graphene makes